

KRANTI DIPAK BHUJBAL

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SUMMARY

Cybersecurity professional with 3+ years of professional experience and hands-on security research experience at SnT, University of Luxembourg. Reverse-engineered 1,000+ malicious APK samples and built a Byzantine-safe security protocol that detected 100% of simulated attacks with zero false positives. Also experienced in IAM and compliance from working in enterprise technology environments. Holds an MSc in Cyber Security & Cyber Defence and available to start immediately.

WORK EXPERIENCE

Research Intern, Fault-Tolerant Security Systems (MSc Thesis Research) Feb 2026 – Aug 2026

CritiX, SnT, University of Luxembourg

- Built a traffic coordination protocol for RSU-controlled areas that kept the system running when a unit failed or was compromised, using adaptive signal control, load balancing, and vehicle rerouting
- Developed **Byzantine-safe failover** that detects and isolates malicious or faulty RSUs without breaking overall system operation
- Tested the protocol against simulated attacks, including drift-based attacks, using **SUMO** and **OMNeT++** across multiple intersections, achieving 100% detection with zero false positives and improved traffic flow during failures

Malicious APK Analyst (Research Assistant)

Apr 2025 – Jul 2025

TruX, SnT, University of Luxembourg

- Reverse-engineered 1000+ malicious APK samples using smali code analysis, extracting IOCs (IPs, domains, hashes) and mapping attacker behavior to the **MITRE ATT&CK** framework
- Cross-validated findings across four tools (**jadx**, **Androguard**, **VirusTotal**, **FlowDroid**) to catch inconsistencies between automated results and confirm true positives before reporting
- Authored formal malware analysis reports documenting attacker techniques, IOCs, and detection recommendations usable directly by **security operations teams**

Senior System Associate

Nov 2023 – Jul 2024

Infosys, India

- Resolved identity and access management cases end-to-end, from initial request through investigation to final resolution, covering firewall changes, VPN provisioning, and access grants via **ServiceNow**, applying **least-privilege** and **role-based access control (RBAC)** principles on every case
- Strengthened **IAM** hygiene across the organization by triaging **Azure** security alerts and running recurring access reviews in **Azure Active Directory**, catching misconfigurations and excess privilege before they became audit findings
- Automated the entire manual access-control workflow by building **PowerShell** and **Bash scripts**, cutting recurring manual errors by an estimated **80%+** and removing hours of manual checking across access-control support cases and infrastructure workflows

Operations Executive

Jul 2021 – Nov 2023

Infosys, India

- Caught SLA risk early by tracking adherence across cross-functional teams in real time, flagging at-risk items and coordinating directly with stakeholders to resolve issues before escalation
- Strengthened organizational **risk and compliance** posture through ad hoc internal control reviews, consistently closing control gaps before they became formal audit findings
- Supported **governance** and reporting for PMO operations, aligning cross-functional activities and maintaining consistent, **audit-ready** reporting to management

EDUCATION

Master in Cyber Security & Cyber Defence, University of Luxembourg

Sep 2024 – Aug 2026

Relevant coursework: Applied Cryptography

Bachelor of Computer Science, GPA: 9.89/10, Fergusson College, Pune, India

Jun 2018 – May 2021

Relevant project: built a library management website using PHP on a Linux-hosted environment

PROJECTS

CVE-2021-3156 (Baron Samedit) – Exploitation & Mitigation

May 2025

- Exploited a heap-based buffer overflow in sudo to gain root access on Ubuntu 20.04.1, then verified the fix on the patched version

MITM Attacks – ARP Spoofing & TLS Enforcement

Dec 2025

- Used ARP spoofing with Bettercap to intercept HTTP credentials, then fixed the issue by enforcing HTTPS/TLS and setting iptables firewall rules

CAFA – Cost-Aware Feasible Attack Path Modelling

Nov 2024 – Feb 2025

- Modelled cost-aware attack scenarios using the CAFA framework; showed targeted cost increases eliminate all viable attack vectors

TECHNICAL SKILLS

Malware & TI: Malware Analysis, Reverse Engineering, IOC Extraction, MITRE ATT&CK, Static & Dynamic Analysis

Security: Network Security, Vulnerability Analysis, **Threat Detection**, **Firewall/VPN**, **Azure IAM**, Log Analysis

Compliance: EU Frameworks (NIS2, ENISA, GDPR), Risk Management, Security Governance

Tools: jadx, Androguard, FlowDroid, VirusTotal, Wireshark, Bettercap, SUMO, OMNeT++, ServiceNow, Azure, Docker, Git

Programming: Python, PowerShell, Bash, Java, JavaScript, PHP, PostgreSQL, MySQL, R

AWARDS & CERTIFICATIONS

2nd Place – CyberEdu4Teens Hackathon | Google.org × SnT, University of Luxembourg | Nov 2025

Designed OopsOnline, a cybersecurity awareness card game for teenagers (12–17), selected for rollout in Luxembourg high schools 2026.

Certifications: SC-900: Microsoft Security · AZ-800 / DP-900: Microsoft Azure · Cybersecurity in SDLC (2022)

Languages: English (C1 – IELTS 7.5) · Marathi (Native) · French (A1) · German (A1)